



Introduction to Time Series Transcript

Hello everyone, welcome to lesson 1 video 1. In this session, we will introduce time series data. Time series analysis is all about understanding how data evolves over time and it uses that knowledge to make forecasts. Hi everyone, welcome to this new module which is called as time series analysis.

In this module, we will be learning more about how do we analyse data which is time dependent. Now, when we say time dependent data, it is about understanding the past and using that understanding that we have developed from the past to forecast about the future. Time series data is a specific type of data which you will learn in this entire module.

There are different types of videos we will be covering but let's get started with it. So, what exactly is a time series data? Time series data is a data which is spaced equally in time. When I say equally spaced in time, it means that it is recorded at equal intervals.

So, let's say if I am recording monthly sales over month over month. So, starting from January to February, March, April, May and then so on. So, that's a monthly sales time series data which is intervalled at months.

Every month you are recording or let's say at every month end or month beginning or mid of the month, you are recording the same monthly sales value and you are doing it for each and every month. So, they need to be equally spaced. When I say months, it means that there is a gap of 30 days.

The same data could be recorded over days. So, you could record on 1st day, 2nd day, 3rd day, 4th day, 5th day, 6th day, 7th day, 8th day and so on. So, every day is gapped at 24 hours.

Similarly, you can record data at every second, every minute and so on. Every year or every 10 years, whatever it is, when you have a data which is spaced equally or it is intervalled equally in time, that is called as a time series data. Now, what are the key characteristics of a time series data? It has two different characteristics.

One is, it is time dependent. What do I mean by time dependent is, today's value is dependent on what has happened in the past. So, past time values or let's say past data values are going to influence my today's value.

So, that is temporal dependence or time dependence which we also call as autocorrelation. Autocorrelation is when we have a value which is getting influenced by its own past value. Let's say for example, monthly sales.

The sales in the month of July is influenced by what it was in the month of June, April, May and so on. Then, that is called as autocorrelation. The second key characteristic is frequency.



Frequency is at what intervals the data is recorded. I have already told you, it could be hourly, daily, monthly, quarterly, annually, whatever. So, these are various different characteristics of the data.

The examples that you see of time series, you could be actually seeing it very oftenly but could not recognise that. Daily stock prices, if you think of, that's a time series data. Monthly sales figures, quarterly company earnings, annual GDP, hours of website traffic and so on.

So, all of that is actually time series data and you can use that data to predict what would happen future in time. So, what do we do with time series data? There are two different types of things. One, understand how the present value is correlated with the past value and what are the implications of it.

That's the first thing. Second, predicting the future because you have understood how the current value is dependent on the past value. So, you can also understand how the past value is going to be dependent on the current set of values.

Hence, you can predict the future. Once you have built the relationship, how the values are dependent or how the entire time series data is dependent on its autocorrelation, you can easily predict the future. That's what is time series analysis.

Now, in the next section, we are going to delve deeper into trend, seasonality, irregularity. We will be decomposing the entire time series into these different components and try to see how each of these components make sense. In summary, time series data arises whenever observations are collected sequentially over time.

Understanding these patterns allows us to build models and that allows us to make predictions about future. Thank you.

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